



DAY 1

WEEK 2 — WEB EXPLOITATION

Web Fundamentals

What is the Web?

The web is built on a client-server model. When you type a URL in your browser, your browser (the client) sends a request to a server, which responds with data — usually HTML, CSS, and JavaScript. As a web hacker, understanding this cycle is your most fundamental skill. Every attack you'll ever do on a web application starts by understanding how data flows between the client and the server.

Key Concepts

HTTP & HTTPS

HTTP (HyperText Transfer Protocol) is the language browsers and servers use to talk to each other. HTTPS is the same but with encryption (TLS/SSL).

HTTP

Sends data in plain text — can be intercepted!

HTTPS

Encrypts data — much safer for sensitive info like passwords.

HTTP Methods

These are the 'verbs' of HTTP — they describe what action the client wants to perform:

Method	Purpose	Example
GET	Retrieve data from server	Load a webpage, fetch an image
POST	Send data to server	Submit a login form
PUT	Update a resource	Edit your profile info
DELETE	Remove a resource	Delete a post

Anatomy of a URL

```
https://www.example.com:443/path/page?id=5&name=test#section
```

scheme	host	port	path	query params	fragment

HTTP Request & Response

Every time you visit a webpage, here's what happens behind the scenes:

— REQUEST (Browser → Server) —————

```
GET /login HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0
Cookie: session=abc123
```

— RESPONSE (Server → Browser) —————

```
HTTP/1.1 200 OK
Content-Type: text/html
Set-Cookie: session=abc123; HttpOnly
```

```
<html>...</html>
```

Common Status Codes

- 200 OK — Request succeeded
- 301/302 — Redirect to another page
- 403 Forbidden — You're not allowed here (but it exists!)
- 404 Not Found — Page doesn't exist
- 500 Internal Server Error — Server crashed (great sign for hackers!)

Tools You'll Use Today

Browser DevTools

Press F12 in any browser → Network tab to see all HTTP requests/responses live.

curl

Command-line tool to make HTTP requests manually. Pre-installed on Linux/Mac.

```
# Make a GET request
curl https://example.com
```

```
# Show response headers
curl -I https://example.com
```

```
# Send a POST request
curl -X POST -d 'username=admin&password=1234'
https://example.com/login
```

```
# Follow redirects
curl -L https://example.com
```

🎯 Your Task for Today

► Flag Challenge — Analyze a Webpage

1. Open your browser and go to: <https://httpbin.org>
2. Open DevTools (F12) → go to the Network tab
3. Visit <https://httpbin.org/get> — observe the request/response
4. Using curl in your terminal, run:
5. `curl -I https://httpbin.org/get`
6. Find and note: the Server header value and the Content-Type
7. Now send a POST request using curl to <https://httpbin.org/post>
8. with data: `name=YourName&club=HackShield`
9. Capture the JSON response — the 'form' field is your flag!
10. BONUS: Try adding a custom header: `curl -H 'X-HackShield: true' https://httpbin.org/headers`

📖 Reference Links

- [MDN — How the Web works \(Beginner\)](#)
- [HTTP Status Codes — Every code explained](#)
- [curl tutorial for beginners](#)
- [PortSwigger Web Academy — Web Fundamentals](#)
- [PicoCTF Web Exploitation Challenges](#)